proposition 3.6.

$$(3.25)\quad F^{\circ}_{k-1}(\theta,\phi) = \omega \frac{k}{\sqrt{8\pi}} P_k^{(0,0)}(\cos\phi) - \omega \frac{1}{\sqrt{8\pi}} P_k^{(0,0)'}(\cos\phi)\sin\phi\, e_{13} e^{-e_{12}\theta}$$

$$F^{\circ}_{k-1} = \partial_x H^{\circ}_k$$

Theorem 3.5 $\Rightarrow$

$$\| F^{\circ}_{k-1} \|^2 = k(2k+3-2) \int_{S^{m-1}} |H^{\circ}_k(\omega)|^2 d\omega \times \frac{1}{2(2k+1)}$$

$$= k(2k+1) \times \frac{1}{2(2k+1)}$$

$$= \frac{k}{2}$$

$$\| H^{\circ}_k(\theta,\phi) \|^2_{L^2(S^2)} = \int_{S^2} \left| \frac{1}{\sqrt{8\pi}} P_k^{(0,0)}(\cos\phi) \right|^2 \sin\phi\, d\omega$$

$\omega \in S^2$

$m=2$

$$= \int_0^{2\pi} \int_0^{\pi} \left| \frac{1}{\sqrt{8\pi}} P_k^{(0,0)}(\cos\phi) \right|^2 \sin\phi\, d\phi\, d\theta$$

$$= \frac{2\pi}{8\pi} \int_0^{\pi} \left| P_k^{(0,0)}(\cos\phi) \right|^2 \sin\phi \, d\phi$$

$\cos\phi = x$

$-\sin\phi \, d\phi = dx$

$\cos 0 = 1$

$\cos\pi = -1$

$$= \frac{1}{4} \int_{-1}^{1} \left| P_k^{(0,0)}(x) \right|^2 dx$$

$$= \frac{1}{4} \int_{-1}^{1} \underbrace{\left| P_k(x) \right|^2}_{\text{Legendre polynomial}} dx = \frac{1}{4} \times \frac{2}{2k+1} = \frac{1}{2(2k+1)}$$

(3.26)

$$I_1 = \frac{(k-n)^2}{2^{2n+1}} \int_0^{\pi} (\sin\phi)^{2n-1} \left(P_{k-n}^{(n,n)}(\cos\phi) \right)^2 d\phi$$

$= 0$

$x = \cos\phi$

$dx = -\sin\phi \, d\phi$

$$\int_{\cos(0)}^{\cos(\pi)} = -\int_{1}^{-1} = \int_{-1}^{+1}$$

$$= \frac{(k-n)^2}{2^{2n+2}} \int_0^{\pi} (\sin^2\phi)^{n-1} \left(P_{k-n}^{(n,n)}(\cos\phi)\right)^2 \sin\phi \, d\phi$$

$$= \frac{(k-n)^2}{2^{2n+2}} \int_{-1}^{+1} (1-x^2)^{n-1} \left| P_{k-n}^{(n,n)}(x) \right|^2 dx$$

$$= \frac{(k-n)^2}{(2^{2n+2})} \int_{-1}^{+1} (1-x^2)^{n-1} (1+x) \left| P_{k-n}^{(n,n)}(x) \right|^2 dx$$

$$- \frac{(k-n)^2}{2^{2n+2}} \int_{-1}^{+1} x \underbrace{(1-x^2)^{n-1}}_{\text{even}} \underbrace{\left| P_{k-n}^{(n,n)}(x) \right|^2}_{\text{even}} dx$$

$P_{k-n}^{(n,n)}(-x) = (-1)^n P_{k-n}^{(n,n)}(x)$

odd $= 0$

$$= \frac{(k-n)^2}{2^{2n+2}} \int_{-1}^{+1} (1-x)^{n-1} (1+x)^n \left| P_{k-n}^{(n,n)}(x) \right|^2 dx$$

$$\overset{\text{eq } 5-7.391}{=} \frac{(k-n)^2}{2^{2n+2}} \times \frac{2^{2n} \Gamma(k+1)^2}{n \; (k-n)! \; \Gamma(k+n+1)}$$

table of integrals

$\alpha = n$

$\beta = n$

$n = k-n$

eq (3.15)

$$I_2 = \frac{k(n-k)}{2^{2n+1}} \int_0^{\pi} (\cos\phi)(\sin\phi)^{2n-2} P_{k-n}^{(n,n)}(\cos\phi)$$

$$P_{k-n-1}^{(n,n)}(\cos\phi) \sin\phi \, d\phi$$

$$= \frac{k(n-k)}{2^{2n+1}} \int_{-1}^{+1} x(1-x^2)^{n-1} P_{k-n}^{(n,n)}(x) P_{k-n-1}^{(n,n)}(x) \, dx$$

by (2.7) we have got

$$(2k-2n+n+n)(1-x^2) P_{k-n}^{(n,n)'}(x)$$

$$= (k-n)\left[(n-n) - (2k-2n+n+n)x\right] P_{k-n}^{(n,n)}(x)$$

$$+ 2(k-n+n)(k-n+n) P_{k-n-1}^{(n,n)}(x)$$

$$\Rightarrow \quad 2k(1-x^2) P_{k-n}^{(n,n)'}(x) = -2k(k-n)x P_{k-n}^{(n,n)}(x)$$

$$+ 2k^2 P_{k-n-1}^{(n,n)}(x)$$

$$\Rightarrow \quad (1-x^2)\, P^{(n,n)\prime}_{k-n}(x) = -(k-n)\, x\, P^{(n,n)}_{k-n}(x) + k\, P^{(n,n)}_{k-n-1}(x)$$

$$\Rightarrow \quad k\, P^{(n,n)}_{k-n-1}(x) = (k-n)\, x\, P^{(n,n)}_{k-n}(x) + (1-x^2) P^{(n,n)\prime}_{k-n}(x)$$

So I_2 can be writen as

$$I_2 = -\frac{(n-k)^2}{2^{2n+1}} \int_{-1}^{+1} x^2 (1-x^2)^{n-1} \left(P^{(n,n)}_{k-n}(x)\right)^2 dx$$

$$+ \frac{(n-k)}{2^{2n+1}} \int_{-1}^{+1} x\,(1-x^2)^n \left(P^{(n,n)}_{k-n}(x)\right)\left(P^{(n,n)\prime}_{k-n}(x)\right) dx$$

Now we try to calculate these two integrals.

$$\int_{-1}^{+1} x^2 (1-x^2)^{m-1} \left(P^{(n,n)}_{k-n}(x)\right)^2 dx$$

$$= -\int_{-1}^{+1} (1-x^2-1)\,(1-x^2)^{n-1}\left(P_{k-n}^{(n,n)}(x)\right)^2 dx$$

$$= -\int_{-1}^{+1} (1-x^2)^n \left(P_{k-n}^{(n,n)}(x)\right)^2 dx + \int_{-1}^{+1} (1-x^2)^{n-1}\left(P_{k-n}^{(n,n)}(x)\right)^2 dx$$

$$= -\frac{2^{2n+1}\,\Gamma(k+1)^2}{(k-n)!\,(2k+1)\,\Gamma(k+n+1)} + \frac{2^{2n}\,\Gamma(k+1)^2}{n\,(k-n)!\,\Gamma(k+n+1)}$$

$$\int_{-1}^{+1} x\,(1-x^2)^n \left(P_{k-n}^{(n,n)}(x)\right)\left(P_{k-n}^{(n,n)}(x)\right)' dx =$$

$$\frac{1}{2}\int_{-1}^{+1} x\,(1-x^2)^n \left(\left(P_{k-n}^{(n,n)}(x)\right)^2\right)' dx$$

by parts

$$= \frac{1}{2}\left[x\,(1-x^2)^n \left(P_{k-n}^{(n,n)}(x)\right)^2\right]\Bigg|_{-1}^{+1}$$

$$-\frac{1}{2}\int_{-1}^{+1} (1-x^2)^n \left(P_{k-n}^{(n,n)}(x)\right)^2 dx$$

$$+\frac{1}{2}\int_{-1}^{+1} 2n\,x^2\,(1-x^2)^{n-1}\left(P_{k-n}^{(n,n)}(x)\right)^2 dx$$

$$= -\frac{1}{2} \int_{-1}^{+1} (1-x^2)^n \left(P_{k-n}^{(n,n)}(x) \right)^2 dx$$

$$+ n \int x^2 (1-x^2)^{n-1} \left(P_{k-n}^{(n,n)}(x) \right)^2 dx$$

$$= -\frac{1}{2} \frac{2^{2n+1} \Gamma(k+1)^2}{(k-n)!\,(2k+1)\,\Gamma(k+n+1)}$$

as before,

$$- \frac{2^{2n+1} \Gamma(k+1)^2 n}{(k-n)!\,(2k+1)\,\Gamma(k+n+1)} \quad \frac{+2^{2n} \Gamma(k+1)^2 \cancel{n}}{\cancel{n}\,(k-n)!\,\Gamma(k+n+1)}$$

$$I_3 = I_2 \cdot$$

$$I_4 = \frac{k^2}{2^{2n+1}} \int_0^{\pi} (\sin\phi)^{2n-1} \left(P_{k-n-1}^{(n,n)}(\cos\phi) \right)^2 d\phi$$

$$= \frac{k^2}{2^{2n+1}} \int_0^{\pi} (\sin\phi)^{2n-2} \underbrace{(1-x^2)^{n-1}}_{\left(P_{k-n-1}^{(n,n)}(x) \right)^2} \left(P_{k-n-1}^{(n,n)}(\cos\phi) \right)^2 \underbrace{\sin\phi\, d\phi}_{dx}$$

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$$= \frac{k^2}{2^{2n+1}} \times \frac{2^{2n}\,\Gamma(k+1-1)^2}{n\,(k-n-1)!_0\,\Gamma(1+k-1+n+1)}$$

$$I_1 = \frac{(k-n)^2}{2^{2n+2}1} \times \frac{2^{2n}\,\Gamma(k+1)^2}{2n\,(k-n)!_0\,\Gamma(1+k+n+1)}$$

$$(k-n-1)!_0$$

$$= \frac{(k!_0)^2\,(k-n)}{2n(k-n-1)!_0\,(k+n)!_0} \checkmark$$

$$I_4 = \frac{k^2\,((k-1)!_0)^2\,(k!_0)^2(k-n)}{2n\,(k-n-1)!_0\,(k+n)!_0} \checkmark$$

$$I_2 = I_3 = -\frac{(n-k)^2}{2^{2n+1}} \times \left[-\frac{2^{2n+1}\,\Gamma(k+1)^2}{(k-n)!_0\,(2k+1)\,\Gamma(k+n+1)} + \frac{2^{2n}\,\Gamma(k+1)^2}{n\,(k-n)!_0\,\Gamma(1+k+n+1)} \right]$$

A B

$$+\frac{(n-k)}{2^{2n+1}}\times\left[-\frac{1}{2}\frac{2^{2n+1}\,\Gamma(k+1)^2}{(k-n)!\,(2k+1)\,\Gamma(k+n+1)}\right.$$

$$\left.-\frac{2^{2n+1}\,\Gamma(k+1)^2\,n}{(k-n)!\,(2k+1)\,\Gamma(k+n+1)}+\frac{2^{2n}\,\Gamma(k+1)^2\,n}{n\,(k-n)!\,\Gamma(k+n+1)}\right]$$

$$=\frac{-(n-k)^2}{2^{2n+1}}\left[-A+\frac{1}{n}B\right]$$

$$+\frac{(n-k)}{2^{2n+1}}\left[\frac{-1}{2k+1}B-nA+B\right]$$

$$=\frac{-(n-k)}{2^{2n+1}}\left[(n-k)A+\frac{(n-k)}{n}B\right]$$

$$=\frac{-(n-k)}{2^{2n+1}}\left[\frac{B}{2k+1}+nA-B\right]$$

$$= \frac{-(n-k)}{2^{2n+1}} \left[-(n-k)A + \frac{(n-k)}{n}B \right.$$

$$\left. + \frac{B}{2k+1} + nA - B \right]$$

$$= \frac{-(n-k)}{2^{2n+1}} \times \left[-nA + kA + B - \frac{k}{n}B \right.$$

$$\left. + \frac{B}{2k+1} + An - B \right]$$

$$= \frac{-(n-k)}{2^{2n+1}} \times \left[\cdot kA \right.$$

$$\left. + \left(\frac{1}{2k+1} - \frac{k}{n} \right) \times B \right]$$

$$= \frac{(n-k)}{2^{2n+1}} \left[-kA + \frac{k}{n}B - \frac{B}{2k+1} \right]$$

From definitions of A and B

we have $A = \frac{2}{2k+1} B$

$$\frac{k}{n} B = \frac{k}{n} \times \frac{2k+1}{2} \times A = \frac{k(2k+1)}{2n} A$$

So

$$I_2 = \frac{(n-k)}{2^{2n+1}} \left[-kA - \frac{A}{2} + \frac{k(2k+1)}{2n} A \right]$$

$$= \frac{(n-k)}{2^{2n+1}} A \left[-k - \frac{1}{2} + \frac{k(2k+1)}{2n} \right]$$

$$\frac{-2kn}{2n} - \frac{2n}{2n} = \frac{-n(2k+1)}{2n}$$

$$= \frac{(n-k)A}{2^{2n+1}} \left[\frac{(k-n)(2k+1)}{2n} \right]$$

$$= \frac{(n-k)}{2^{2n+1}} \left[\frac{(k-n)}{n} B \right]$$

$$= \frac{-(k-n)^2}{n\, 2^{2n+1}} \beta$$

$$\Rightarrow I_2 = I_3 = \frac{-(k-n)^2}{n\, 2^{2n+1}} \times \frac{2^{2n}\, \Gamma(k+1)^2}{(k-n)!\, \Gamma(k+n+1)}$$

$$(k-n-1)!$$

$$= \frac{-(k-n)\, \Gamma(k+1)^2}{2n\, (k-n-1)!\, (k+n)!}$$

Now, $I_1 + I_2 + I_3 + I_4$

$$= \frac{(k-n)(k!)^2}{2n\, (k-n-1)!\, (k+n)!}$$

$$- 2\, \frac{(k-n)\, (k!)^2}{2n\, (k-n-1)!\, (k+n)!}$$

$$+ \frac{(k+n)\, (k!)^2}{2n\, (k-n-1)!\, (k+n)!}$$

$$= \frac{(k!_0)^2 (k+n)}{2n\,(k-n-1)!_0\,(k+n)!_0} - \frac{(k-n)\,(k!_0)^2}{2n\,(k-n-1)!_0\,(k+n)!_0}$$

$$= \frac{(k!_0)^2 \cancel{2n}}{\cancel{2n}\,(k-n-1)!_0\,(k+n)!_0}$$

$$= \frac{(k!_0)^2}{(k-n-1)!_0\,(k+n)!_0}$$